

OP AMP AND ITS CHARACTERISTICS

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INTRODUCTION TO OP-AMPS

- An **Operational Amplifier (Op-Amp)** is a highly versatile electronic component used for signal amplification, mathematical operations, and various analog processing applications. It consists of a high-gain differential amplifier with two inputs (inverting and non-inverting) and one output.

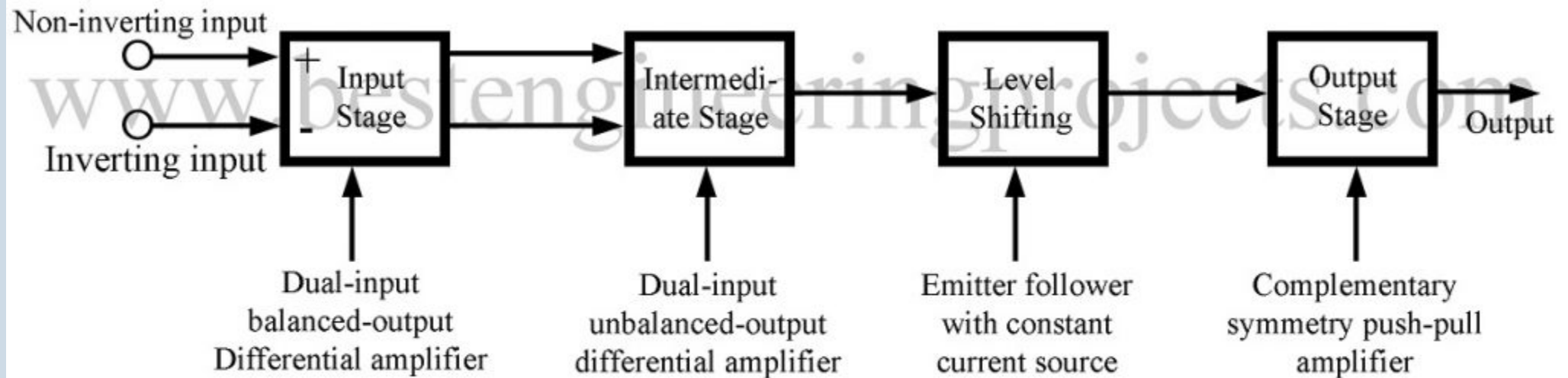


Figure 1: Block Diagram of a Typical Op-amp

INTERNAL STRUCTURE OF OP-AMP

- A typical Op-Amp consists of three major stages:

1. Differential Input Stage:

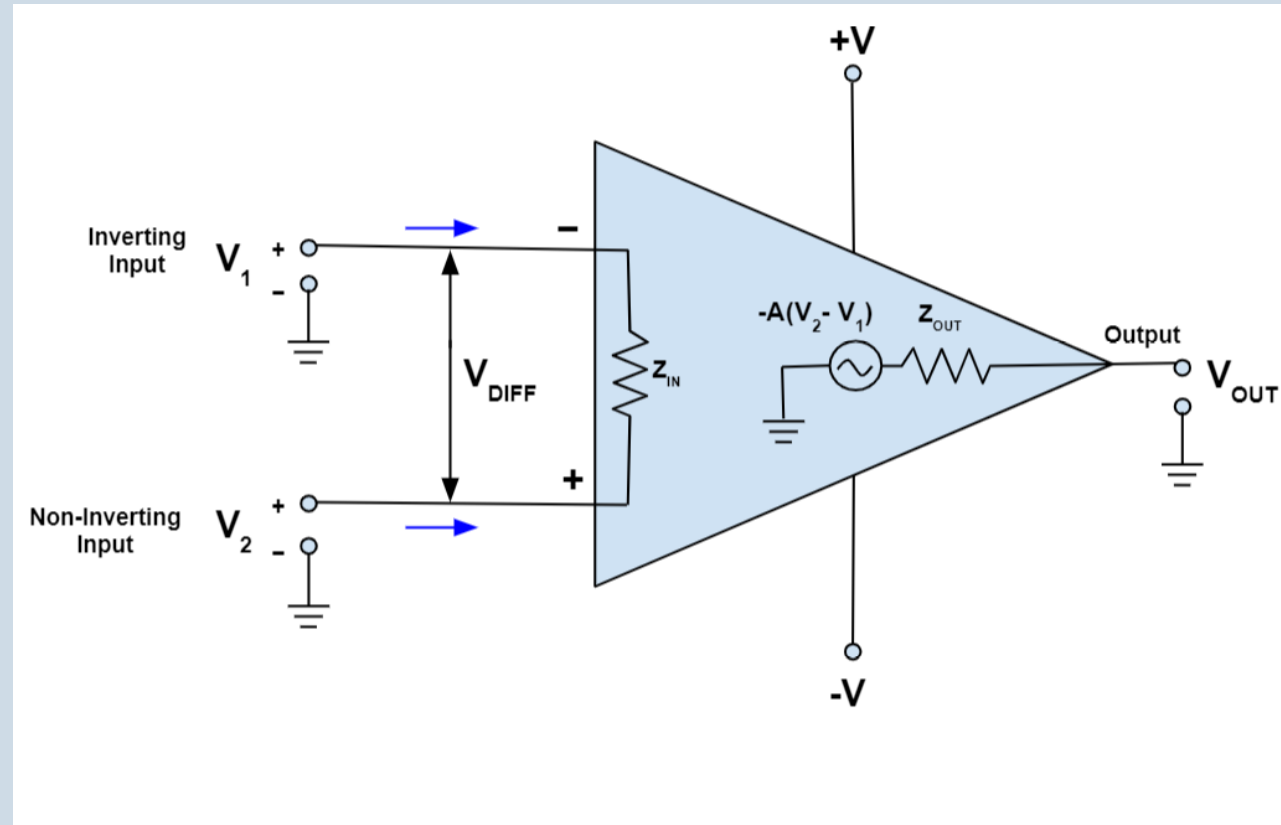
1. Composed of a differential amplifier with current mirrors.
2. Ensures high common-mode rejection and initial gain.

2. Gain Stage (Intermediate Stage):

1. Amplifies the signal further.
2. Improves the voltage gain of the Op-Amp.

3. Output Stage:

1. Buffers the signal to drive external loads.
2. Provides low output impedance to maintain signal integrity.



IDEAL VS PRACTICAL OP-AMP

- **Ideal Op-Amp:**

- Infinite input impedance
- Zero output impedance
- Infinite open-loop gain
- Infinite bandwidth
- Zero noise contribution
- Zero DC output offset

- **Practical Op-Amp:**

- Input impedance: $500\text{k}\Omega$ to $2\text{M}\Omega$
- Output impedance: 20Ω to 100Ω
- Open-loop gain: 20,000 to 200,000
- Bandwidth is limited (a few kHz)
- Has noise contribution
- Non-zero DC output offset

OP-AMP CONFIGURATION

- Op-Amps can be used in different configurations based on feedback:

1. Inverting Amplifier:

1. Provides phase-inverted output.
2. Gain = $-R_f/R_{in}$

2. Non-Inverting Amplifier:

1. Output in phase with input.
2. Gain = $1 + R_f/R_{in}$

3. Voltage Follower (Buffer):

1. Gain = 1
2. Used for impedance matching.

OP-AMP PARAMETERS & PERFORMANCE METRICS

- Some key performance parameters that define an Op-Amp's efficiency:
 - **Open-Loop Gain (A_{OL}):** Typically very high ($\sim 10^5$).
 - **Input & Output Impedance:** Should be high and low, respectively.
 - **CMRR (Common-Mode Rejection Ratio):** Should be high for noise rejection.
 - **Slew Rate:** Speed at which output changes; higher is better for fast signals.
 - **Bandwidth:** Frequency range where Op-Amp operates effectively.
- These parameters influence the Op-Amp's stability, accuracy, and application scope.

FREQUENCY RESPONSE & STABILITY

- Op-Amps exhibit different gain at different frequencies:
 - **At Low Frequencies:** Gain remains constant.
 - **At High Frequencies:** Gain decreases due to internal capacitance.
 - **Gain-Bandwidth Product (GBW):** Constant for a given Op-Amp.
 - **Compensation Techniques:**
To maintain stability in high-frequency applications:
 - **Miller Compensation:** Uses an internal capacitor to stabilize feedback.
 - **Pole-Splitting:** Separates poles in the frequency response to prevent oscillations.

I-STAGE OP AMP DESIGN

Model or Device Parameters	Value
V_{tp}	-0.46 V
V_{tn}	0.49 V
K'_p	63 $\mu\text{A}/\text{V}^2$
K'_n	325 $\mu\text{A}/\text{V}^2$

Step1:- To meet the slew rate, and maximum Power Dissipation

$$I_5 = \text{SR} \cdot C_L$$

$$P_{diss} = (V_{DD} + |V_{SS}|) \cdot I_5$$

Step2:- Design for M1 and M2 to achieve desired GB.

$$g_{m1,2} = \text{GB} \times C_L$$
$$\left(\frac{W}{L}\right)_{1,2} = \frac{g_m^2}{2I_D K'_n}$$

Step3:- Design for M3 and M4 from the maximum input voltage [ICMR(+)] specification.

$$\left(\frac{W}{L}\right)_{3,4} = \frac{2I_3}{K'_p \{V_{DD} - \text{ICMR}(+) - V_{T3\max} + V_{T1\min}\}^2}$$

Step4:- Design for M5 and M6 from the minimum input voltage [ICMR(-)]. First we have calculated $V_{DS5(sat)}$ and then we have find M5 and M6.

$$V_{DS5(sat)} = ICMR(-) - \left(\frac{I_5}{\beta_1}\right)^{\frac{1}{2}} + V_{T1max}$$

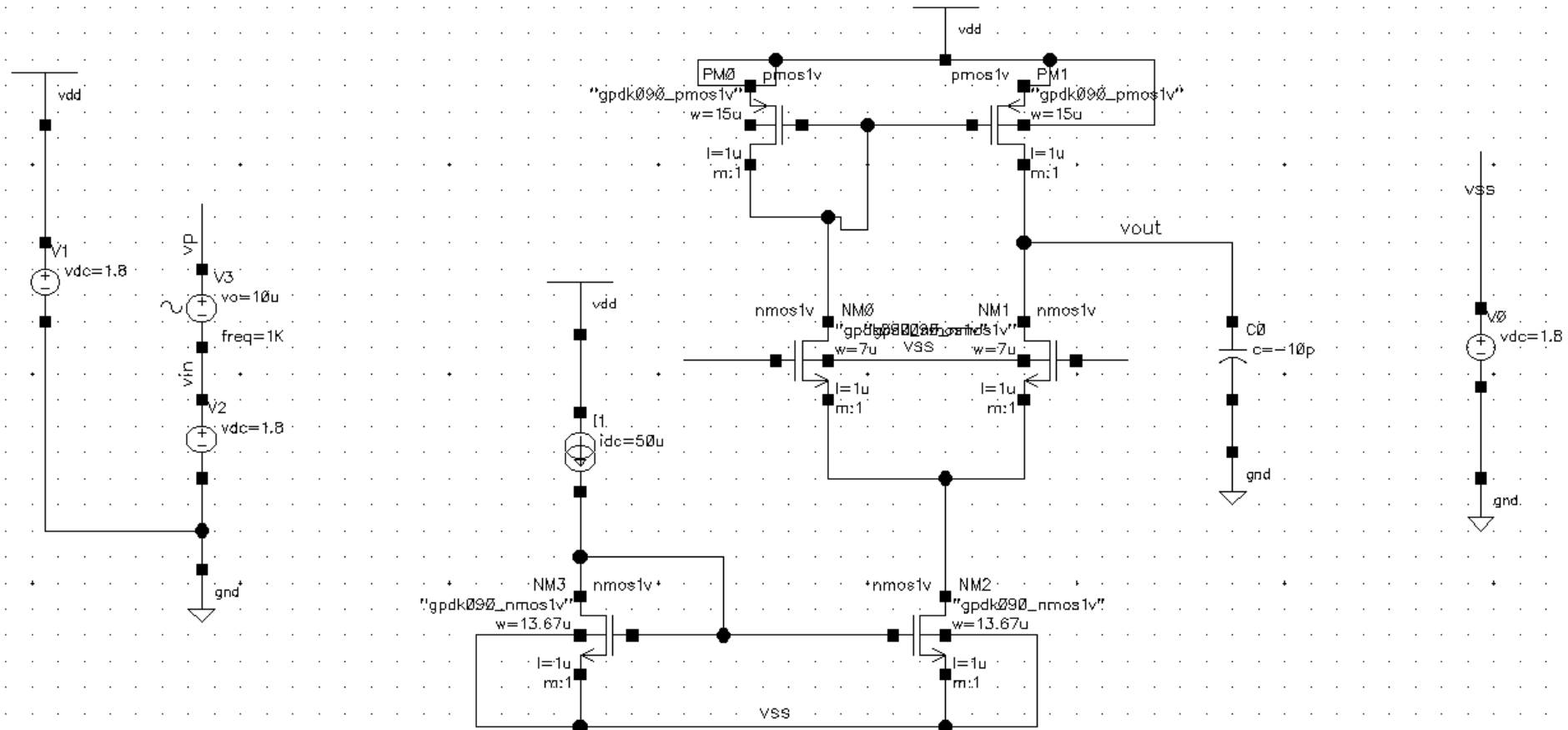
$$\left(\frac{W}{L}\right)_{5,6} = \frac{2I_5}{K'_n[V_{DSsat}]^2}$$

DESIGN SPECIFICATION

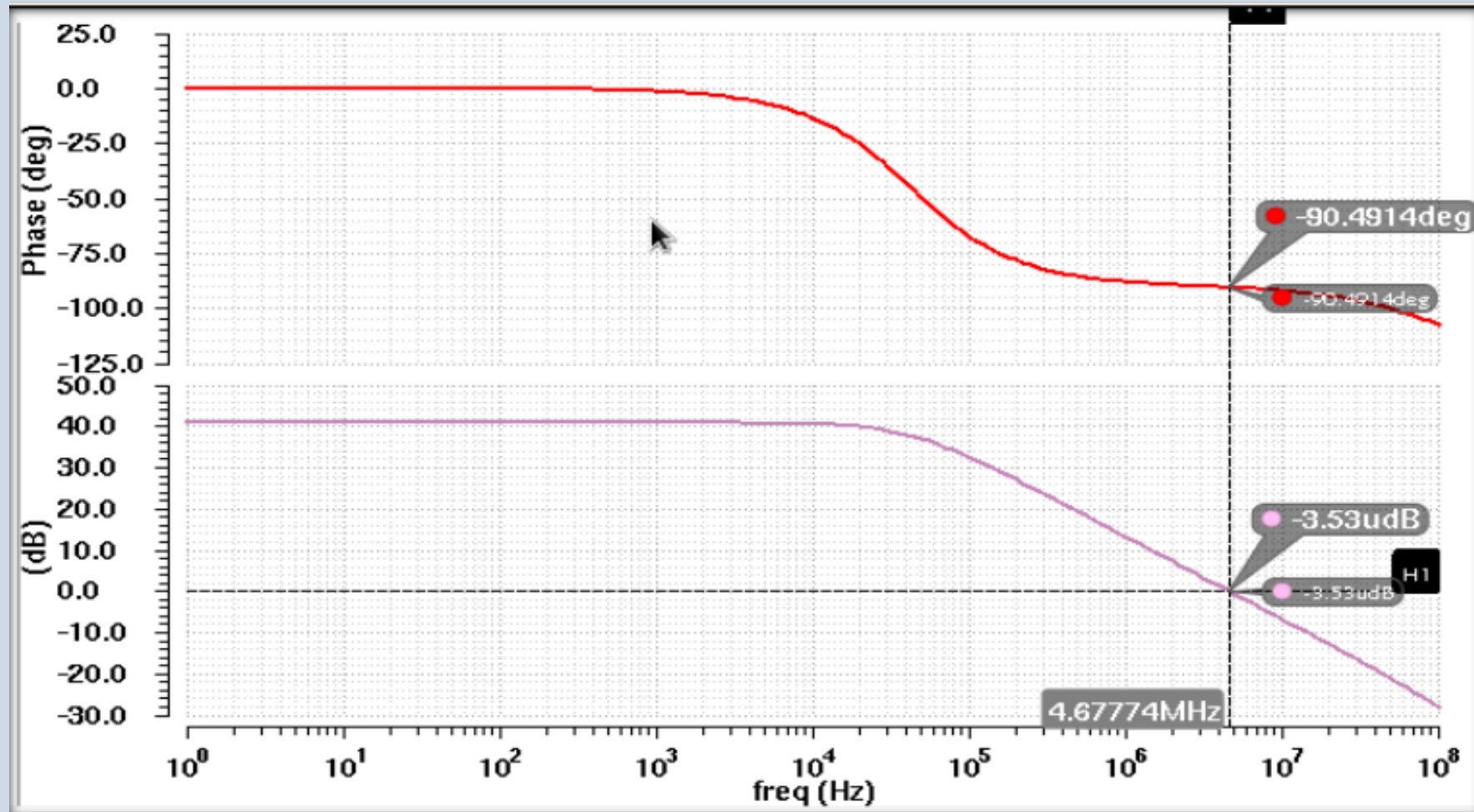
Sl. No.	Specifications	Value
1.	Technology	gpdk180
2.	Power Supply (V_{DD})	1.8 V
3.	Load Capacitance (C_L)	10 pF
4.	Gain	≥ 40 dB
5.	Gain Bandwidth	≥ 5 MHz
6.	Slew rate (SR)	5 V/usec

MOSFETS	(W/L)
M1	7
M2	7
M3	15
M4	15
M5	13.67
M6	13.67

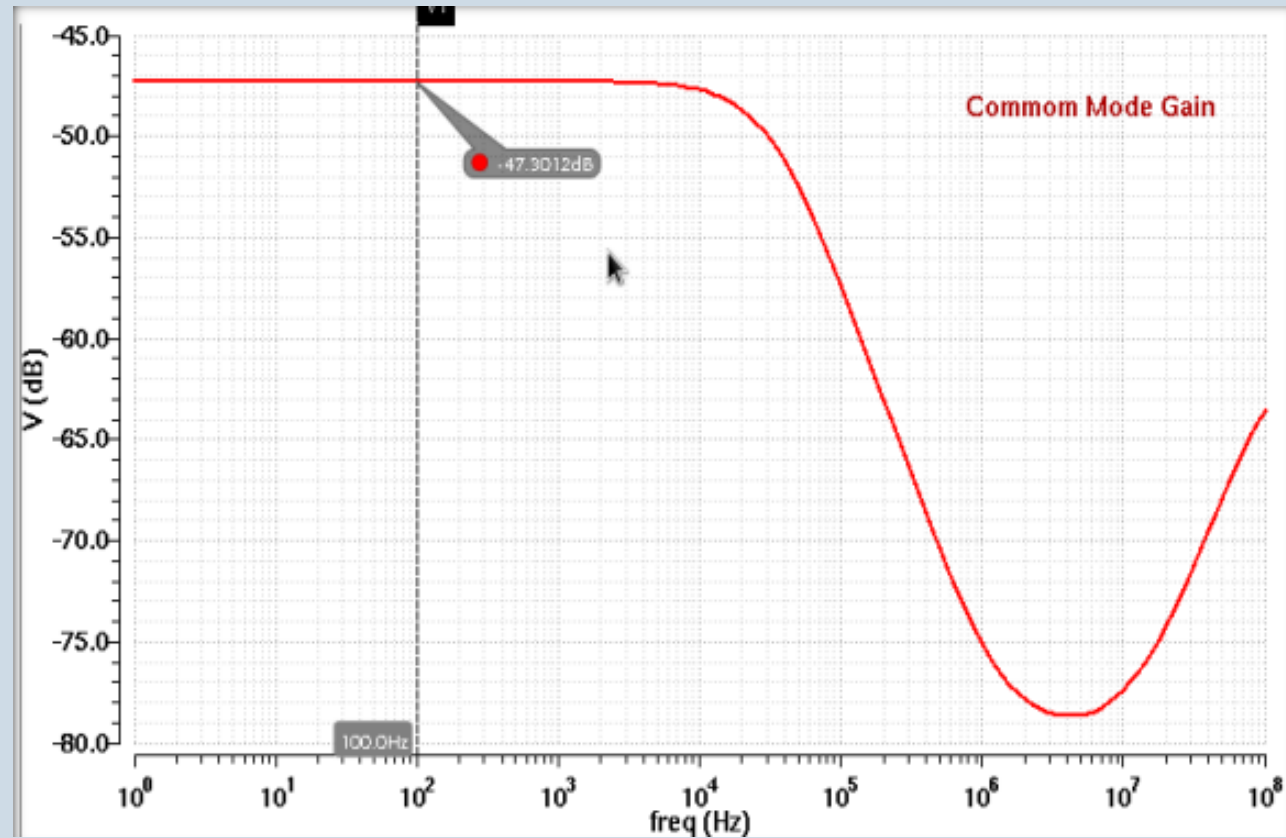
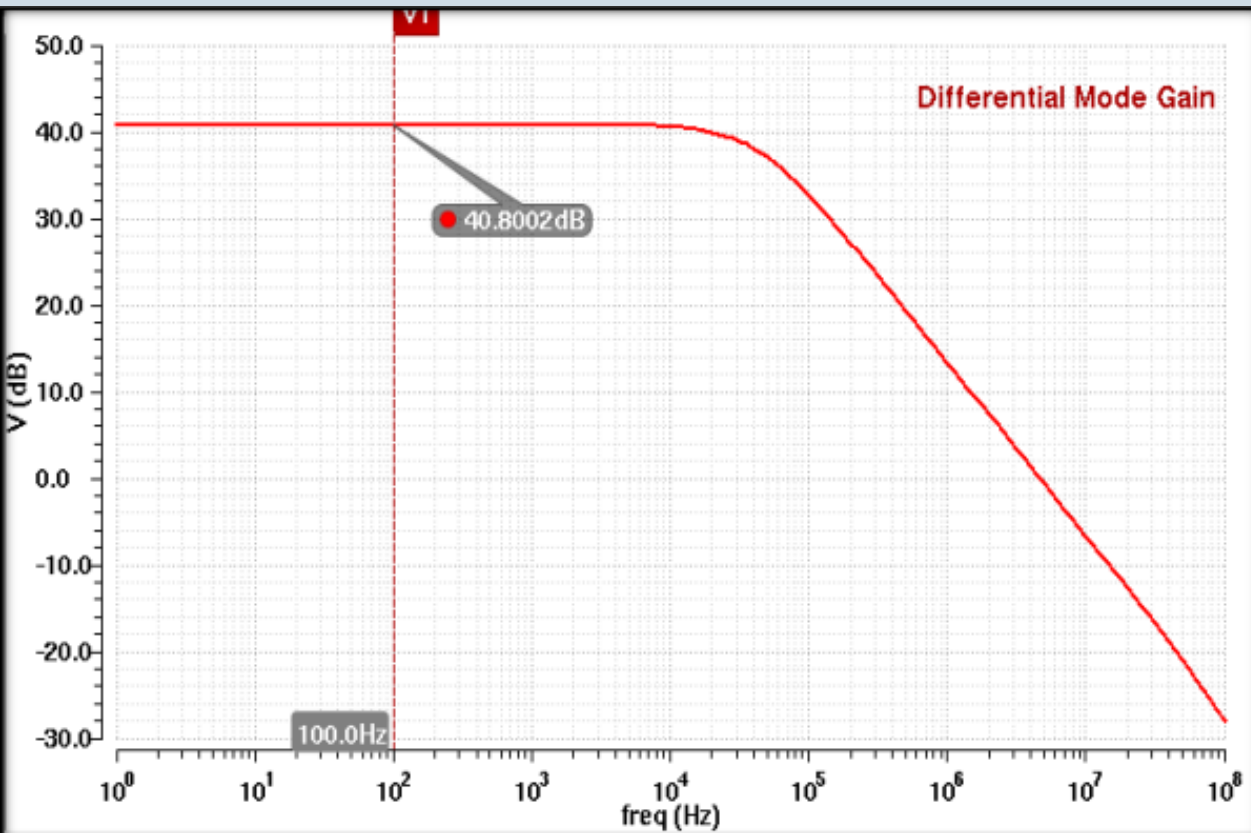
I-STAGE OP-AMP SCHEMATIC



GAIN AND PHASE



DIFFERENTIAL & COMMON MODE GAIN



CMRR & SLEW RATE

$CMRR = AD - (-AC)$ in dB $AD = 40.8002$ dB and $AC = -47.3012$ dB.

So, $CMRR = 88.10$ dB

Slew Rate

Slew rate = 5.32 V/us

2-OP AMP DESIGN

1. From the desired phase margin, the minimum value for C_c is chosen. We have used the following relationship. This assumes that $\omega \geq 10GB$.

$$\text{For } 45^\circ \text{ phase margin } C_c \geq 0.122C_L$$

$$\text{For } 60^\circ \text{ phase margin } C_c \geq 0.22C_L$$

2. Finding of bias current from the slew rate and compensation capacitor

$$I_5 = SR \cdot C_c$$

3. Design for transistor M1 and M2 to achieve desired GB.

$$g_{m1} = GB \times C_c$$

$$\left(\frac{W}{L}\right)_{1,2} = \frac{g_{m1}^2}{K'_n I_5}$$

4. Design for M3 and M4 from the maximum input voltage [ICMR(+)] specification.

$$\left(\frac{W}{L}\right)_{3,4} = \frac{2I_3}{K'_p \{V_{DD} - ICMR(+) - V_{T3max} + V_{T1min}\}^2}$$

5. Design for M5 from the minimum input voltage [ICMR(-)]. First we have calculated $V_{DS5(sat)}$ and then we have find M5.

$$V_{DS5(sat)} = ICMR(-) - V_{SS} - \left(\frac{I_5}{\beta_1}\right)^{\frac{1}{2}} - V_{T1max}$$

$$\left(\frac{W}{L}\right)_{5,6} = \frac{2I_5}{K'_n[V_{DSsat}]^2}$$

6. Find M6 and I_6 by letting the second pole (p_2) be equal to 2.2 times GB.

$$g_{m6} = 2.2g_{m2} \left(\frac{C_L}{C_C}\right)$$

$$\text{Let } V_{GS4} = V_{GS6}, \text{ which gives } M6 = M4 \left(\frac{g_{m6}}{g_{m4}}\right)$$

$$\text{Knowing } g_{m6} \text{ and M6 allows us to solve for } I_6 \text{ as } I_6 = \frac{g_{m6}^2}{2K'_p M6}$$

7. Calculating of M7 from the M5 , I_6 and I_5

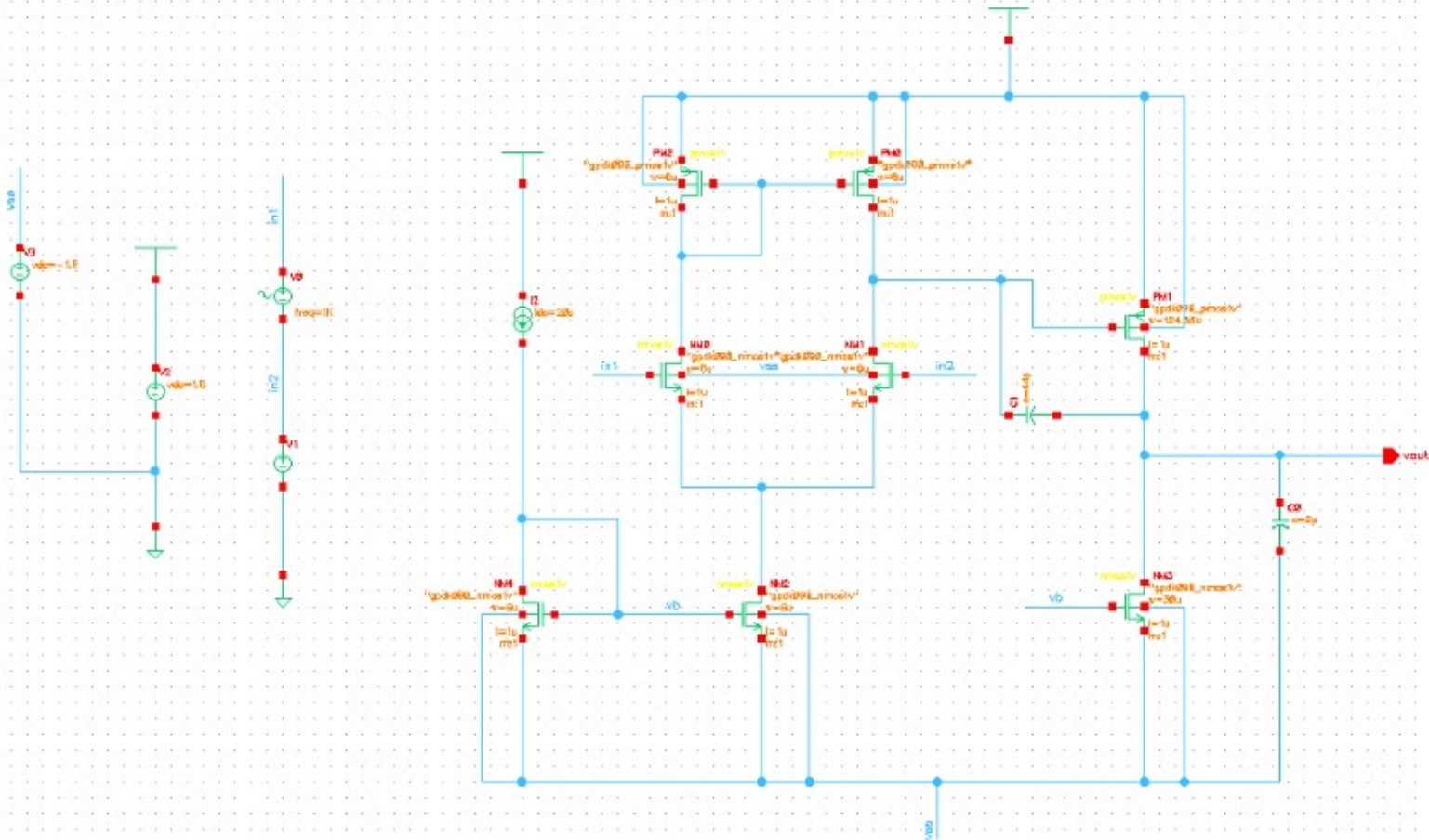
$$M7 = \left(\frac{I_5}{I_6}\right)M6$$

DESIGN SPECIFICATION

Sl. No.	Specifications	Value
1.	Technology	gpdk180
2.	Power Supply (V_{DD})	1.8 V
3.	Load Capacitance (C_L)	2 pF
4.	Gain	60 dB
5.	Phase Margin	≥ 45 degree
6.	Gain Bandwidth	30 MHz
7.	Slew rate (SR)	≥ 42 V/usec

MOSFETS	(W/L)
M1	6
M2	6
M3	8
M4	8
M5	124.56
M6	31.14
M7	4
M8	4

2-OP AMP SCHEMATIC

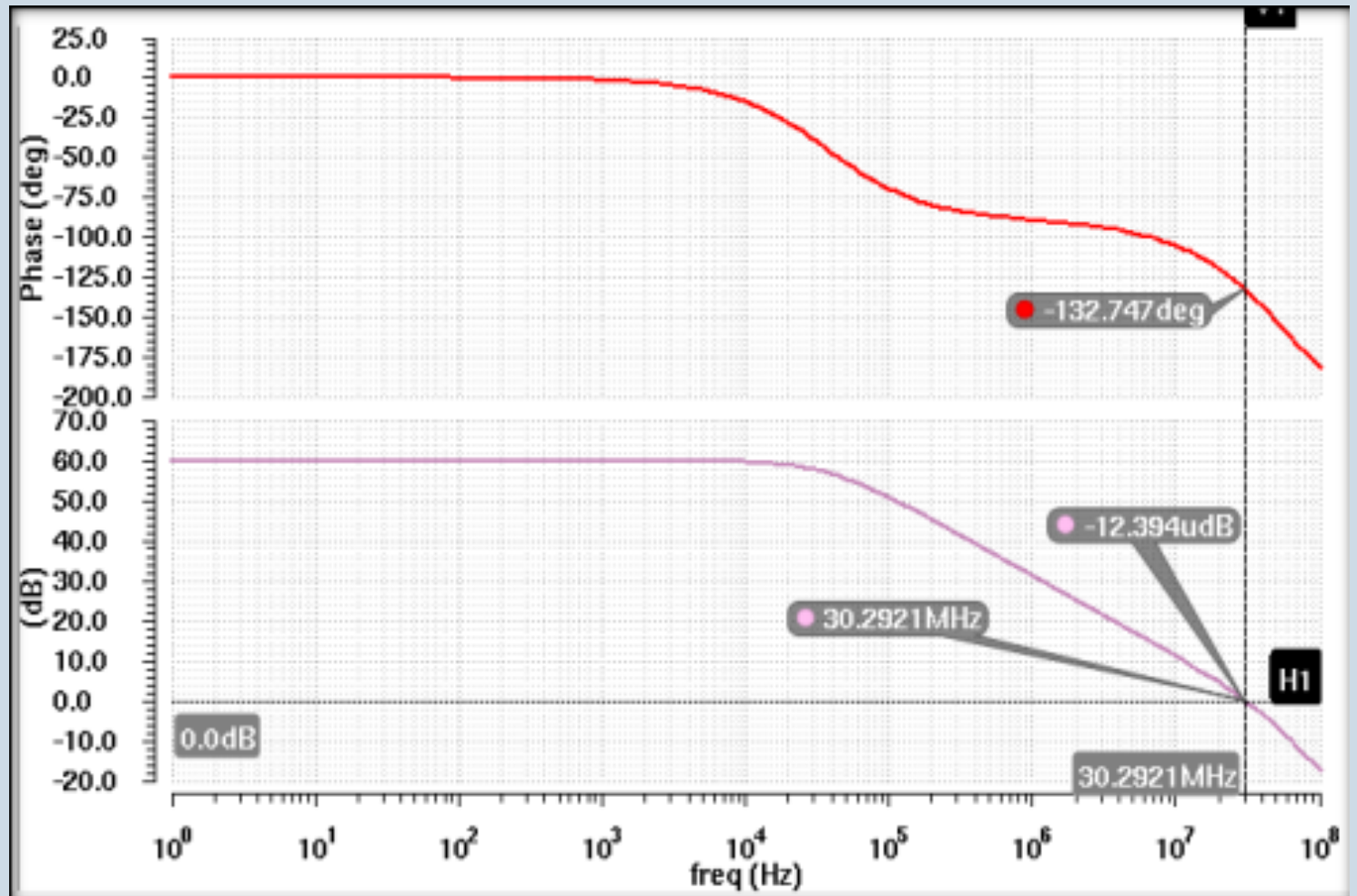


GAIN & PHASE

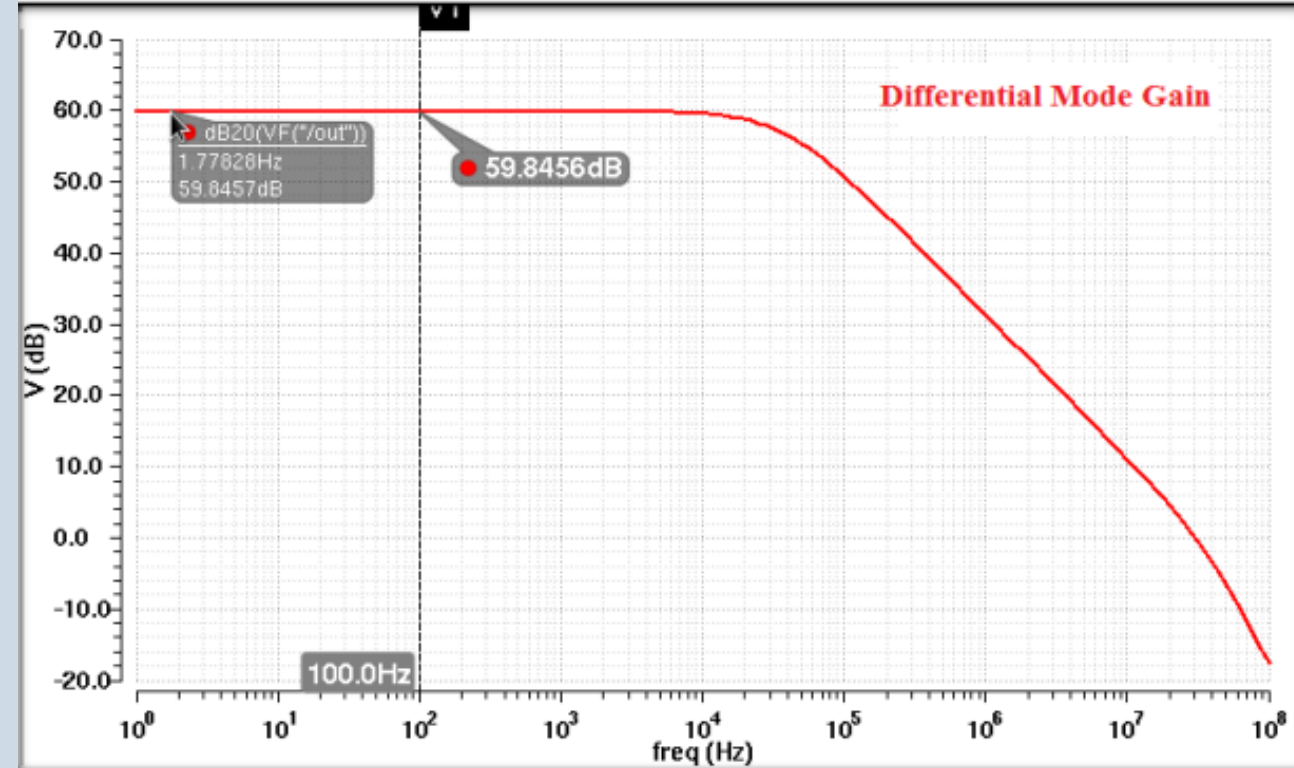
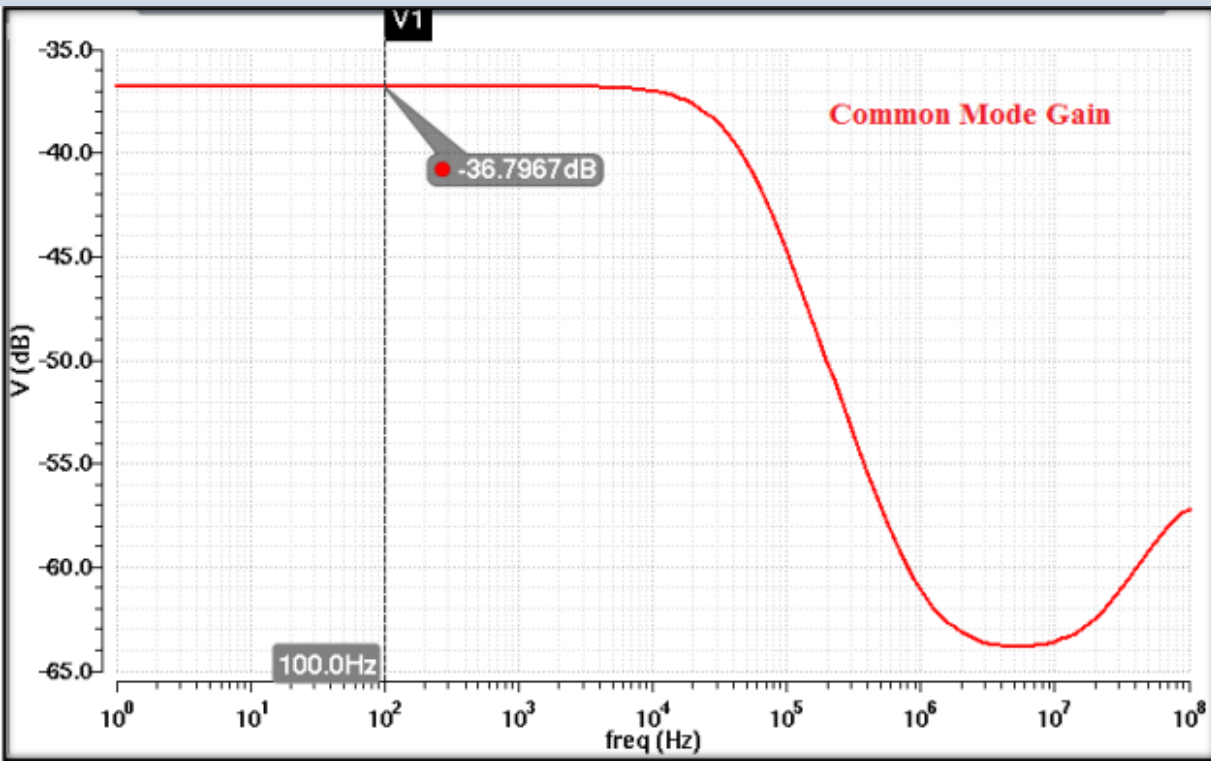
Phase Margin = $180 - 132.747$

= 47.253 degree

Gain Margin = $0 - (-16.8) = 16.8$ dB



DIFFERENTIAL & COMMON MODE GAIN



CMRR & SLEW RATE

$CMRR = AD - (-AC)$ in dB

$AD = 59.8456$ dB and $AC = -36.7967$ dB

So, $CMRR = 96.6423$ dB

Slew rate = 42.5 V/us

CONCLUSION

- **Higher Gain** – Two-stage op-amp (59.85 dB) offers better amplification than single-stage (40.8 dB).
- **Wider Bandwidth** – Two-stage design (30.3 MHz) has a superior frequency response over single-stage (4.69 MHz).
- **Stability Trade-off** – Single-stage has better stability (89.5° phase margin) compared to two-stage (47.253°).
- **Better Performance** – Two-stage op-amp excels in CMRR and Slew Rate for precision applications.

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